



Report 6

Mid-Term Assignment

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Company Scenario:

TechNova Solutions is deploying a new local area network (LAN) for a small enterprise branch. The objective of this project is to design, subnet, configure, and verify a scalable network using VLSM, VLANs, static routing, and routing isolation techniques. The network is designed to support four departments: Sales, IT, Finance, and HR.

Objectives:

- Design an enterprise network using Cisco Packet Tracer.
- Perform VLSM subnetting.
- Configure VLANs for each department.
- Configure static routing between routers.
- Implement routing isolation.
- Verify network connectivity using ping tests.

Task 1: VLSM Subnetting

Base Network: 192.168.10.0/24

Available IPs = 256

Arrange Departments (Largest to Smallest):

Department	Hosts Needed
Sales	60
IT	30
Finance	14
HR	6

VLSM Prefix Requirements:

Department	Required Hosts	Required Addresses	CIDR Prefix	Subnet Mask
Sales	60	64	/26	255.255.255.192
IT	30	32	/27	255.255.255.224
Finance	14	16	/28	255.255.255.240
HR	6	6	/29	255.255.255.248

VLSM Table:

Department	Network Address	Prefix	Subnet Mask	First Usable	Last Usable	Broadcast
Sales	192.168.10.0	/26	255.255.255.192	192.168.10.1	192.168.10.62	192.168.10.63
IT	192.168.10.64	/27	255.255.255.224	192.168.10.65	192.168.10.94	192.168.10.95
Finance	192.168.10.96	/28	255.255.255.240	192.168.10.97	192.168.10.110	192.168.10.111
HR	192.168.10.112	/29	255.255.255.248	192.168.10.113	192.168.10.118	192.168.10.119

Remaining address space:

192.168.10.120 – 192.168.10.255

Can be used later for the Remote Branch

Task 2: Network Architecture

Core Layer

Purpose

- Performs routing
- Connects enterprise networks
- Connects Remote Branch

Device

- Main Router

Distribution Layer

Purpose

- Performs VLAN aggregation
- Trunking
- Policy enforcement

Device

- Managed Switch (2960)

Access Layer

Purpose

- Connects end devices

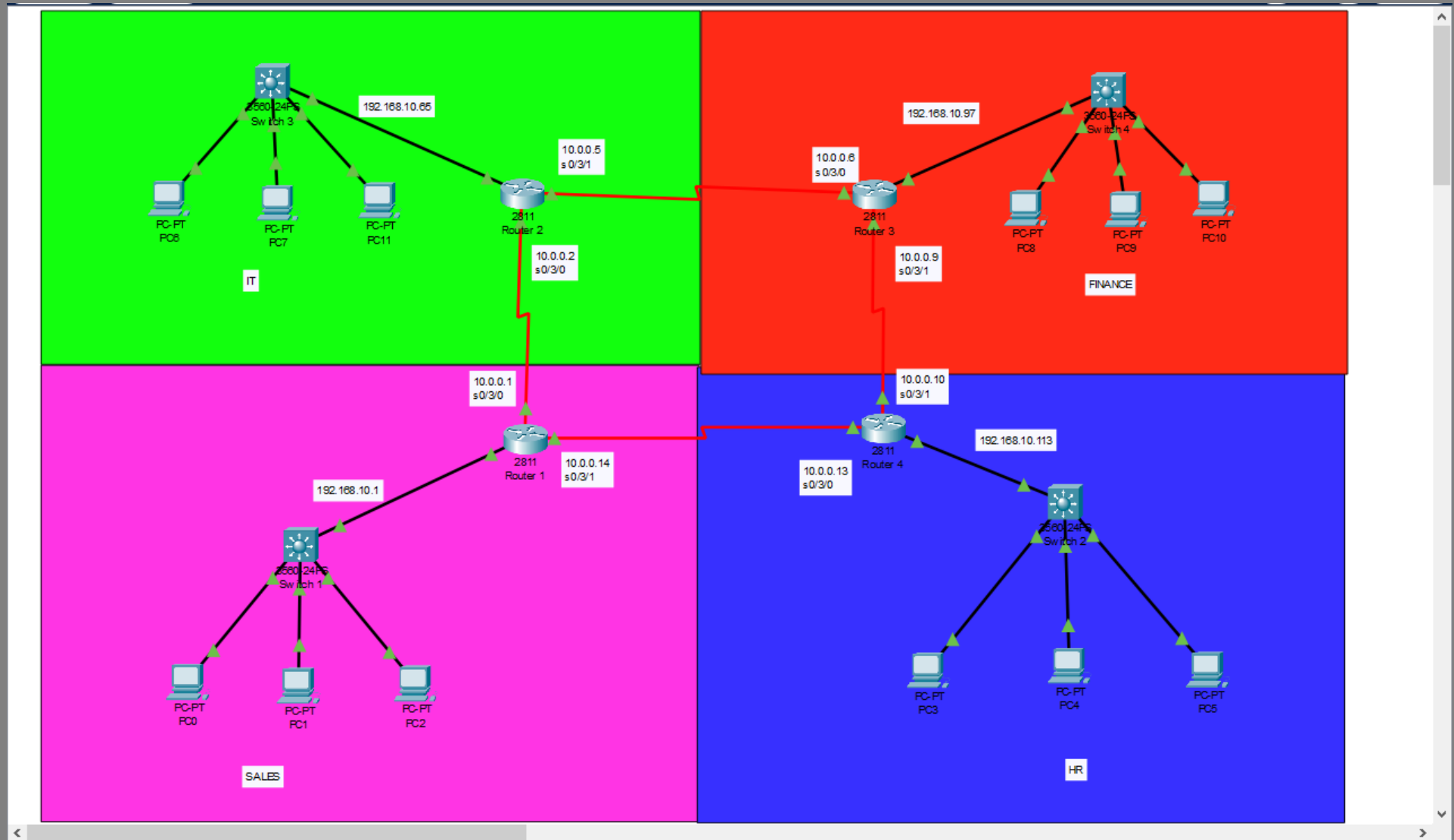
Devices

- PCs
- Printer (if added)

VLAN Planning:

VLAN	Department
Vlan 10	Sales
Vlan 20	IT
Vlan 30	Finance
Vlan 40	HR

Topology:



Router 1:

Show ip int brief:

```
R1#
R1#
R1#show ip int brief
Interface                IP-Address      OK? Method Status              Protocol
FastEthernet0/0          192.168.10.1    YES manual up                    up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/3/0               10.0.0.1        YES manual up                    up
Serial0/3/1               10.0.0.14       YES manual up                    up
Vlan1                     unassigned      YES unset  administratively down down
R1#
R1#
R1#
R1#
R1#
R1#
R1#
```

Show ip Route:

```
R1#
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
C       10.0.0.2/32 is directly connected, Serial0/3/0
C       10.0.0.12/30 is directly connected, Serial0/3/1
C       10.0.0.13/32 is directly connected, Serial0/3/1
L       10.0.0.14/32 is directly connected, Serial0/3/1
    192.168.10.0/24 is variably subnetted, 5 subnets, 5 masks
C       192.168.10.0/26 is directly connected, FastEthernet0/0
L       192.168.10.1/32 is directly connected, FastEthernet0/0
S       192.168.10.64/27 [1/0] via 10.0.0.2
S       192.168.10.96/28 [1/0] via 10.0.0.2
S       192.168.10.112/29 [1/0] via 10.0.0.13
R1#
R1#
R1#
```

Show running-config:

```
!
interface FastEthernet0/0
 ip address 192.168.10.1 255.255.255.192
 duplex auto
 speed auto
!
interface FastEthernet0/1
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Serial0/3/0
 bandwidth 1000
 ip address 10.0.0.1 255.255.255.252
 encapsulation ppp
!
interface Serial0/3/1
 bandwidth 1000
 ip address 10.0.0.14 255.255.255.252
 encapsulation ppp
!
interface Vlan1
 no ip address
 shutdown
!
ip classless
ip route 192.168.10.64 255.255.255.224 10.0.0.2
ip route 192.168.10.96 255.255.255.240 10.0.0.2
ip route 192.168.10.112 255.255.255.248 10.0.0.13
!
ip flow-export version 9
!
!
!
```

Router 2:

Show ip int brief:

```
R2>
R2>en
R2#
R2#show ip int brief
Interface          IP-Address      OK? Method Status              Protocol
FastEthernet0/0    192.168.10.65   YES manual up                up
FastEthernet0/1    unassigned      YES unset  administratively down down
Serial0/3/0        10.0.0.2        YES manual up                up
Serial0/3/1        10.0.0.5        YES manual up                up
Vlan1              unassigned      YES unset  administratively down down
R2#
```

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Show Ip route:

```
Vlan1              unassigned      YES unset  administratively down down
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
C       10.0.0.1/32 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
C       10.0.0.4/30 is directly connected, Serial0/3/1
L       10.0.0.5/32 is directly connected, Serial0/3/1
C       10.0.0.6/32 is directly connected, Serial0/3/1
S       192.168.10.0/24 is variably subnetted, 5 subnets, 5 masks
S       192.168.10.0/26 [1/0] via 10.0.0.1
C       192.168.10.64/27 is directly connected, FastEthernet0/0
L       192.168.10.65/32 is directly connected, FastEthernet0/0
S       192.168.10.96/28 [1/0] via 10.0.0.6
S       192.168.10.112/29 [1/0] via 10.0.0.6
```

Show running-config:

```
!
interface FastEthernet0/0
ip address 192.168.10.65 255.255.255.224
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial10/3/0
bandwidth 1000
ip address 10.0.0.2 255.255.255.252
encapsulation ppp
clock rate 64000
!
interface Serial10/3/1
bandwidth 1000
ip address 10.0.0.5 255.255.255.252
encapsulation ppp
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.10.0 255.255.255.192 10.0.0.1
ip route 192.168.10.96 255.255.255.240 10.0.0.6
ip route 192.168.10.112 255.255.255.248 10.0.0.6
!
ip flow-export version 9
!
```

Router 3:

Show ip int brief:

```
[OK]
R3#
R3#
R3#
R3#
R3#show ip int brief
Interface      IP-Address      OK? Method Status      Protocol
FastEthernet0/0 192.168.10.97   YES manual up          up
FastEthernet0/1 unassigned      YES unset  administratively down down
Serial10/3/0    10.0.0.6        YES manual up          up
Serial10/3/1    10.0.0.9        YES manual up          up
Vlan1           unassigned      YES unset  administratively down down
R3#
```

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Show ip route:

```
Vlan1          unassigned      YES unset  administratively down down
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C       10.0.0.4/30 is directly connected, Serial0/3/0
C       10.0.0.5/32 is directly connected, Serial0/3/0
L       10.0.0.6/32 is directly connected, Serial0/3/0
C       10.0.0.8/30 is directly connected, Serial0/3/1
L       10.0.0.9/32 is directly connected, Serial0/3/1
C       10.0.0.10/32 is directly connected, Serial0/3/1
    192.168.10.0/24 is variably subnetted, 5 subnets, 5 masks
S       192.168.10.0/26 [1/0] via 10.0.0.5
S       192.168.10.64/27 [1/0] via 10.0.0.5
C       192.168.10.96/28 is directly connected, FastEthernet0/0
L       192.168.10.97/32 is directly connected, FastEthernet0/0
S       192.168.10.112/29 [1/0] via 10.0.0.10
```

Show running-config:

```
!
interface FastEthernet0/0
ip address 192.168.10.97 255.255.255.240
ip access-group 100 in
duplex auto
speed auto
!
interface FastEthernet0/1
no ip address
duplex auto
speed auto
shutdown
!
interface Serial0/3/0
bandwidth 1000
ip address 10.0.0.6 255.255.255.252
encapsulation ppp
clock rate 64000
!
interface Serial0/3/1
bandwidth 1000
ip address 10.0.0.9 255.255.255.252
encapsulation ppp
!
interface Vlan1
no ip address
shutdown
!
ip classless
ip route 192.168.10.0 255.255.255.192 10.0.0.5
ip route 192.168.10.64 255.255.255.224 10.0.0.5
ip route 192.168.10.112 255.255.255.248 10.0.0.10
!
ip flow-export version 9
!
!
access-list 100 deny ip 192.168.10.96 0.0.0.15 192.168.10.0 0.0.0.63
access-list 100 permit ip any any
!
!
```

Show access-list router 3 (Finance):

```
R3#  
R3#  
R3#  
R3#  
R3#show access-list  
Extended IP access list 100  
 10 deny ip 192.168.10.96 0.0.0.15 192.168.10.0 0.0.0.63 (4 match(es))  
 20 permit ip any any (4 match(es))  
  
R3#D
```

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Router 4:

Show ip int brief:

```
[OK]  
R4#  
R4#  
R4#  
R4#show ip int brief  
Interface                IP-Address      OK? Method Status          Protocol  
FastEthernet0/0          192.168.10.113  YES manual  up              up  
FastEthernet0/1          unassigned      YES unset    administratively down down  
Serial0/3/0              10.0.0.13       YES manual  up              up  
Serial0/3/1              10.0.0.10       YES manual  up              up  
Vlan1                    unassigned      YES unset    administratively down down  
R4#
```

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Show ip Route:

```
R4#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/8 is variably subnetted, 6 subnets, 2 masks
C    10.0.0.8/30 is directly connected, Serial0/3/1
C    10.0.0.9/32 is directly connected, Serial0/3/1
L    10.0.0.10/32 is directly connected, Serial0/3/1
C    10.0.0.12/30 is directly connected, Serial0/3/0
L    10.0.0.13/32 is directly connected, Serial0/3/0
C    10.0.0.14/32 is directly connected, Serial0/3/0
192.168.10.0/24 is variably subnetted, 5 subnets, 5 masks
S    192.168.10.0/26 [1/0] via 10.0.0.14
S    192.168.10.64/27 [1/0] via 10.0.0.9
S    192.168.10.96/28 [1/0] via 10.0.0.9
C    192.168.10.112/29 is directly connected, FastEthernet0/0
L    192.168.10.113/32 is directly connected, FastEthernet0/0
```

Show Access-lists:

```
R4#
R4#
R4#show access-lists
Extended IP access list 101
 10 deny ip 192.168.10.112 0.0.0.7 192.168.10.0 0.0.0.63 (4 match(es))
 20 permit ip any any (16 match(es))
```

R4#

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Show running-config:

```
!  
!  
interface FastEthernet0/0  
 ip address 192.168.10.113 255.255.255.248  
 ip access-group 101 in  
 duplex auto  
 speed auto  
!  
interface FastEthernet0/1  
 no ip address  
 duplex auto  
 speed auto  
 shutdown  
!  
interface Serial0/3/0  
 bandwidth 1000  
 ip address 10.0.0.13 255.255.255.252  
 encapsulation ppp  
 clock rate 64000  
!  
interface Serial0/3/1  
 bandwidth 1000  
 ip address 10.0.0.10 255.255.255.252  
 encapsulation ppp  
 clock rate 64000  
!  
interface Vlan1  
 no ip address  
 shutdown  
!  
ip classless  
ip route 192.168.10.0 255.255.255.192 10.0.0.14  
ip route 192.168.10.64 255.255.255.224 10.0.0.9  
ip route 192.168.10.96 255.255.255.240 10.0.0.9  
!  
ip flow-export version 9  
!  
!  
access-list 101 deny ip 192.168.10.112 0.0.0.7 192.168.10.0 0.0.0.63  
access-list 101 permit ip any any  
!  
!  
!
```

Sales To HR ping test:

```
C:\>ping 192.168.10.113

Pinging 192.168.10.113 with 32 bytes of data:

Reply from 192.168.10.113: bytes=32 time=1ms TTL=254
Reply from 192.168.10.113: bytes=32 time=28ms TTL=254
Reply from 192.168.10.113: bytes=32 time=19ms TTL=254
Reply from 192.168.10.113: bytes=32 time=1ms TTL=254

Ping statistics for 192.168.10.113:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 28ms, Average = 12ms

C:\>
```

HR to Sales ping test:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.113: Destination host unreachable.
Reply from 192.168.10.113: Destination host unreachable.
Reply from 192.168.10.113: Destination host unreachable.
Reply from 192.168.10.113: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Ping failed due to ACL list Applied

IT To Finance Ping:

```
C:\>ping 192.168.10.98

Pinging 192.168.10.98 with 32 bytes of data:

Reply from 192.168.10.98: bytes=32 time=1ms TTL=126
Reply from 192.168.10.98: bytes=32 time=20ms TTL=126
Reply from 192.168.10.98: bytes=32 time=30ms TTL=126
Reply from 192.168.10.98: bytes=32 time=19ms TTL=126

Ping statistics for 192.168.10.98:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 30ms, Average = 17ms

C:\>
```

Finance to Sales ping test:

```
C:\>ping 192.168.10.2

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.
Reply from 192.168.10.97: Destination host unreachable.

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Unreachable ACL Worked Correctly

SW1:

```
SW1#
```

```
SW1#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
10	SALES	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW1#
```

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SW2:

```
SW2>
SW2>en
SW2#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
40	HR	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW2#
```

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SW3:

```
SW3#
```

```
SW3#
```

```
SW3#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
20	IT	active	Fa0/1, Fa0/2, Fa0/3
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

```
SW3#
```

SW4:

```
SW4#  
SW4#  
SW4#show vlan brief
```

VLAN Name	Status	Ports
1 default	active	Fa0/4, Fa0/5, Fa0/6, Fa0/7 Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
30 FINANCE	active	Fa0/1, Fa0/2, Fa0/3
1002 fddi-default	active	
1003 token-ring-default	active	
1004 fddinet-default	active	
1005 trnet-default	active	

```
SW4#
```

Challenges Faced:

During the implementation, ACLs initially allowed communication between the HR and Sales departments because they were applied in the wrong direction. The issue was resolved by applying the ACLs inbound (in) on the LAN interfaces. After correcting the ACL direction, the required communication restrictions were successfully enforced.

Learning Outcomes:

During this assignment, I gained practical experience in designing and implementing an enterprise network using Cisco Packet Tracer. I successfully applied VLSM subnetting to allocate IP addresses efficiently, configured VLANs to segment departmental traffic, implemented static routing between multiple routers, and enforced network access restrictions using ACLs. This project improved my understanding of enterprise network design, routing, switching, and troubleshooting techniques.

Conclusion:

The enterprise network was successfully designed and configured according to the project requirements. VLSM subnetting, VLAN implementation, static routing, and access control were completed successfully. Network connectivity was verified through ping tests, and routing isolation policies were enforced to meet departmental communication requirements. The final topology provides a secure, organized, and scalable network suitable for a small enterprise environment.